F70LA/F71AF – Life Insurance Mathematics 1
Chapter 7 – Policy Values

Read: Dickson, Hardy & Waters, Chapter 7

Policy Values – Motivation:

Example 1 *On 1 January 1990 a life aged 40 purchased a whole life insurance with sum insured £100,000 payable at the end of the year of death and annual premiums payable throughout the term. The premium basis is:*

Tables!
- *Survival model: AM92 Ultimate*
- *Interest: 6% per year*

Expenses: 50% of the first premium and 10% of all subsequent premiums. £100 on payment of the sum insured.

(a) *Calculate the annual premium.*

$$P\,\ddot{a}_{40} = (10^5 + 100)A_{40} + 0.1P\ddot{a}_{40} + 0.4P \rightarrow P = £910.16$$

$$\text{EPV[Profit]} = 0 \rightarrow \text{EPV[Premiums]} = \text{EPV[Benefits + Expenses]}$$

(b) *At 1 January 2005, how much money does the life office **need** in respect of this policy?*

The office needs nothing IF (40) is no longer alive.
Suppose (40) is still alive at age 55.
Calculate the EPV at 1/1/05 of future cashflow. Premium basis?

$$\begin{aligned}\text{EPV[Future benefits + expenses]} &= (10^5 + 100)\,A_{55} + 0.1P\,\ddot{a}_{55} \\ &= £27,306 \\ \text{EPV[Future premiums]} &= P\,\ddot{a}_{55} = £11,884\end{aligned}$$

The office needs $27,306 - 11,884 = £15,422$ in cash on 1/1/05 to balance its books.

(c) *Calculate the corresponding premiums for a 15-year term insurance from age 40 and for a whole life insurance from age 55. (We will need $\ddot{a}_{40:\overline{15}|}$ and $A^{1}_{40:\overline{15}|}$.)*

First,
$\ddot{a}_{40:\overline{15}|} = \ddot{a}_{40} - v^{15}\,\frac{l_{55}}{l_{40}}\,\ddot{a}_{55} = 10.208$
Next,
$A_{40:\overline{15}|} = 1 - d\,\ddot{a}_{40:\overline{15}|}$
And
$A^{1}_{40:\overline{15}|} = v^{15}\,\frac{l_{55}}{l_{40}}$
So
$A^{1}_{40:\overline{15}|} = A_{40:\overline{15}|} - A_{40:\overline{15}|} = 0.01757$

$$\text{Term: } P_T\,\ddot{a}_{40:\overline{15}|} = (10^5 + 100)\,A^{1}_{40:\overline{15}|} + 0.1P_T\,\ddot{a}_{40:\overline{15}|} + 0.4P_T$$
$$\Longrightarrow P_T = £199.97$$
$$\text{WL: } P_{WL}\,\ddot{a}_{55} = (10^5 + 100)\,A_{55} + 0.1P_{WL}\,\ddot{a}_{55} + 0.4P_{WL}$$
$$\Longrightarrow P_{WL} = £2,300.89$$

Conclusions:
$910.16 > 199.97 \Longrightarrow$ The premiums in the first 15 years are more than sufficient to pay for the benefits. $\Longrightarrow$ The extra amount can be saved to provide the cash needed at age 55.
$910.16 < 2,300.89 \Longrightarrow$ From age 55, future premiums are not sufficient to pay for the benefits. $\Longrightarrow$ Draw down on the savings from earlier years to make up the difference.

Important points:

- When a group of policies is issued to lives aged x, the (future) premiums are expected to be sufficient to provide the (future) (benefits + expenses).
- At any later time, t, after the policies have been issued, the **future** premiums may not be expected to provide the **future** (benefits + expenses) for the policies still in force.
- The difference should be provided by the accumulation to time t of the **past** premiums less any **past** (benefits + expenses) paid.
- The amount the office **needs** at time t is the **policy value** (sometimes known as the **reserve** or **prospective policy value**).
- There are two types of policy value: gross and net premium policy values.
- ${}_tV$ is the notation for a policy value at **time** t. $\equiv$ Duration since issue

Policy Values – Definitions:

The **gross premium policy value** per policy in force at time t, is calculated as follows:

$$ {}_tV = \mathrm{E}_t[PV_t(\text{future benefits} + \text{future expenses} - \text{future premiums})] $$

where E_t is the expected value **given that the policyholder is alive at time** t and PV_t is the present value **at time** t**.**

- The premium used in this calculation is the **actual** premium payable by the policyholder
- Care is needed with premiums, benefits and expenses payable **at** time t. Conventionally, a premium or annuity payment payable at time t is a **future** payment and a death benefit payable **at** time t is a **past** benefit.
- The numerical value of the policy value depends on the basis used in its calculation. } Need not be the same as the premium basis
- For with profits policies, the policy value should allow for **actual** bonuses already declared and for future bonuses as assumed in the policy value basis.

The **net premium policy value** per policy in force at time t, is calculated as follows:

$$ {}_tV \;=\; \mathrm{E}_t[\mathrm{PV}_t(\text{future benefits} - \text{future premiums})] \qquad \text{No expenses!} $$

where E_t is the expected value **given that the policyholder is alive at time t** and PV_t is the present value **at time t.**

- The EPVs are calculated on a specified basis (the net premium policy value basis). } Need not be the same as the premium basis
- The premium used in this calculation is the **net premium calculated on the net premium policy value basis**, which may not be the premium payable by the policyholder.
- A net premium policy value is a special case of a gross premium policy value: No expenses and premium calculated on the (net premium) policy value basis.

Example 2 *Calculate the following policy values for the policy in* Example 1 *at 1 January 2007 just before the premium then due is paid and also just after the premium is paid.*

(a) The gross premium policy value using the premium basis.

(b) The net premium policy value using the basis:
Survival model: AM92 Ultimate
Interest 4% per year. ← Example 1: Interest = 6%

Solution:

(a)

$$\begin{aligned} {}_{17}V(\text{before prem.}) &= \left(10^5 + 100\right) A_{57} \;+\; 0.9P\; \ddot{a}_{57} \\ &= £18,312 \\ {}_{17}V(\text{after prem.}) &= \left(10^5 + 100\right) A_{57} \;+\; 0.9P\; a_{57} \\ &= £19,131 \\ {}_{17}V(\text{after}) &= {}_{17}V(\text{before}) \;+\; 0.9P \end{aligned}$$

(b)

$$P^* = \text{net premium:}\quad P^*\,\ddot{a}_{40} = 10^5 A_{40} \;\rightarrow\; P^* = £1,152.51 \neq 910.16$$

$$\left.\begin{aligned} {}_{17}V(\text{before}) &= 10^5 A_{57} - P^*\,\ddot{a}_{57} = £24,044 \\ {}_{17}V(\text{after}) &= 10^5 A_{57} - P^*\,a_{57} = £25,196 \end{aligned}\right\} \text{Differences is } P^*$$

Example 3 *A life aged* 35 *purchases a* 20*−year deferred annuity of* £1,000 *per year to be paid continuously from age* 55*. Premiums are payable continuously for at most* 20 *years.*

(a) Calculate the annual amount of the premium and the gross premium policy value at durations 10*,* 20 *and* 30 *years. The premium and gross premium policy value basis is:*

premium basis = policy value basis (need not be so)

Survival model: AM92 Select

Interest: 6% *per year*

Expenses: Initial expenses £200*, renewal expenses of* 5% *of all premiums.*

(b) Calculate the net premium policy value at durations 10*,* 20 *and* 30 *years using AM92 Select,* 4% *interest.*

Solution:

(a) Gross annual premium is P where AM92 Select at 6%

$$P\,\bar{a}_{[35]:\overline{20}|} = 1{,}000\,A_{[35]:\overline{20}|}^{\,1}\,\bar{a}_{55} + 200 + 0.05\,P\,\bar{a}_{[35]:\overline{20}|}$$

$$\bar{a}_{[35]:\overline{20}|} \overset{\text{WOOLHOUSE}}{\approx} \ddot{a}_{[35]:\overline{20}|} - \frac{1}{2}\left[1 - A_{[35]:\overline{20}|}^{\,1}\right]$$

$$\ddot{a}_{[35]} - A_{[35]:\overline{20}|}^{\,1}\,\ddot{a}_{55} \qquad \frac{l_{55}}{l_{[35]}}v^{20}$$

$$\Longrightarrow P = £358.00$$

$$_{10}V = 1{,}000\,A_{45:\overline{10}|}^{\,1}\,\bar{a}_{55} - 0.95\,P\,\bar{a}_{45:\overline{10}|} = £4{,}282.59$$

(7.51243)

$$_{20}V = 1{,}000\,\bar{a}_{55} = £12{,}557$$

$$_{30}V = 1{,}000\,\bar{a}_{65} = £10{,}069$$

(b) Policy value basis $\neq$ Premium basis, it is AM92 Select at 4%. Net premium is P^* where

$$P^*\,\bar{a}_{[35]:\overline{20}|} = 1{,}000\,A_{[35]:\overline{20}|}^{\,1}\,\bar{a}_{55}$$

$$\Longrightarrow P^* = £493.77$$

$$_{10}V = 1{,}000\,A_{45:\overline{10}|}^{\,1}\,\bar{a}_{55} - P^*\,\bar{a}_{45:\overline{10}|} = £6{,}081$$

$$_{20}V = 1{,}000\,\bar{a}_{55} = £15{,}373$$

$$_{30}V = 1{,}000\,\bar{a}_{65} = £11{,}776$$

Example 4

A life office sells a $10-$*year with profits endowment policy to a life aged exactly* 60.

The basic sum insured of £100,000 *is payable at the end of the year of death or on survival to age* 70.

Level premiums are payable annually in advance throughout the term of the policy.

Compound reversionary bonuses are declared at the start of each year, including the first.

Premium Basis:

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$, *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075.$$

Interest: 6% *per year*

Bonuses: 2% *per year compound reversionary bonuses.*

Assumed bonus is part of the basis. No expenses

Calculate the annual premium.

Solution:

$$\begin{aligned} P\,\ddot{a}_{60:\overline{10}|} &= \text{EPV[Benefits]} \\ &= 10^5 \left[1.02\, v\, q_{60} + 1.02^2\, v^2\, {}_{1|}q_{60} + \cdots \right. \\ &\quad \left. \cdots + 1.02^{10}\, v^{10}\, {}_{9|}q_{60} + 1.02^{10}\, v^{10}\, {}_{10}p_{60}\right] \\ \Longrightarrow \quad P &= £10,540.09 \end{aligned}$$

Ch7Ex48.pdf

The life office wants to calculate the gross premium policy value for this insurance just before the 6th premium is paid and the bonus for the coming year is declared.
The bonus additions made to date amount to £13,000.
The gross premium policy value basis is:

Same as premium basis

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$, *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075.$$

Higher then premium basis

Interest: 8% *per year*

Future bonus additions: 2.5% *per year compound*

Calculate the gross premium policy value.

Solution: The 6th premium is due duration 5, age 65.

$$10^5 \times 1.02^5 = 110,408 < 113,000$$

$\Rightarrow$ Actual bonuses in the first 5 years have been greater than assumed in the premium basis. Probable reason? Policy has made more profit than was expected at time 0.

$$\begin{aligned} {}_5V &= 113,000\left[1.025\, v\, q_{65} + 1.025^2\, v^2\, {}_{1|}q_{65} + \cdots \right. \\ &\quad \left. \cdots + 1.025^5\, v^5\, {}_{4|}q_{65} + 1.025^5\, v^5\, {}_5p_{65}\right] \\ &\quad - 10,540.09\, \ddot{a}_{65:\overline{5}|} \\ &= £46,940.18 \end{aligned}$$

Recursion formulae for policy values:

Example 5

Consider a whole life insurance with sum insured £100,000, payable at the end of the year of death, issued to a life aged 40. Premiums are payable annually throughout life.

The basis for calculating the premium and gross premium policy values is:

Tables! {
- *Mortality: AM92 Select* At age 40
- *Interest: 4% per year*

Expenses: 5% of each gross premium and £100 when the sum insured is paid

$$0.95\ P\ \ddot{a}_{[40]} \ = \ (10^5 \ + \ 100)\ A_{[40]}$$

(a) *Calculate the gross premium, P.* $\rightarrow \ P \ = \ £1,213.35$

(b) *Calculate the gross premium policy value at duration 20, just before the premium then due is paid.* ${}_{20}V \ = \ 100,100 A_{60} - 0.95\ P\ \ddot{a}_{60} \ = \ £29,394$

(c) *Calculate the gross premium policy value at duration 21, just before the premium then due is paid.* ${}_{21}V \ = \ 100,100 A_{61} - 0.95\ P\ \ddot{a}_{61} \ = \ £31,217$

(d) *Show that for any non-negative integer duration, t,*

$$({}_tV + 0.95\ P)(1+i) \ = \ 100,100\ q_{[40]+t} \ + \ p_{[40]+t}\ {}_{t+1}V$$

Solution:

Check for $t \ = \ 20$

$$\left.\begin{aligned} ({}_{20}V \ + \ 0.95\ P)\,(1.04) \ &= 31,768 \\ 100\,100\ q_{60} \ + \ 31,217\ p_{60} \ &= 31,769 \end{aligned}\right\} \text{ OK, Rounding!}$$

In general, ${}_tV \ = \ 100,100\ A_{[40]+t} \ - \ 0.95\ P\ \ddot{a}_{[40]+t}$

$${}_tV \ = \ 100,100\left[v\ q_{[40]+t} \ + \ v\ p_{[40]+t}\ A_{[40]+t+1}\right] \ - \ 0.95\ P\left[1 \ + \ v\ p_{[40]+t}\ \ddot{a}_{[40]+t+1}\right]$$

$$\Rightarrow \ {}_tV + 0.95\ P \ = \ v\left[100,100\left(q_{[40]+t} \ + \ p_{[40]+t}\ A_{[40]+t+1}\right) \ - \ 0.95\ P\ p_{[40]+t}\ \ddot{a}_{[40]+t+1}\right]$$

$$\Rightarrow \ ({}_tV + 0.95\ P)\,(1+i) \ = \ 100,100\ q_{[40]+t} \ + \ p_{[40]+t}\ {}_{t+1}V$$

Policies with discrete (annual) cash flows:

Consider a policy issued to a life (x) where cash flows – premiums, expenses and claims – can occur only at the **start** or **end** of a year. Suppose this policy has been in force for t years, where t is a non-negative integer. Consider the year $t \to t+1$ and let:

P_t denote the premium payable at time t,

e_t denote the premium-related expense payable at time t,

S_{t+1} denote the sum insured payable at time $t+1$ if the policyholder dies in the year,

E_{t+1} denote the expense of paying the sum insured at time $t+1$,

${}_tV$ denote the policy value for a policy in force at time t, and,

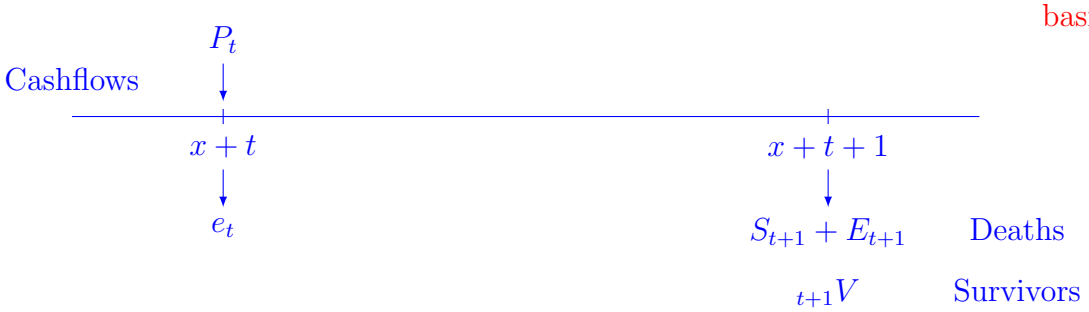
${}_{t+1}V$ denote the policy value for a policy in force at time $t+1$.

Let q_{x+t} denote the probability that the policy holder, alive at time t, dies in the year and let i_t denote the rate of interest assumed earned in the year. The quantities e_t, E_t, q_{x+t} and i_t are all as assumed in the **policy value basis**.

Then:

Not necessarily the same as the premium basis.

$$({}_tV + P_t - e_t)(1 + i_t) = q_{x+t}(S_{t+1} + E_{t+1}) + p_{x+t}\,{}_{t+1}V$$



${}_tV$ = EPV$_t$[All cashflows from $t \to$] ← Assuming alive at time t

${}_{t+1}V$ = EPV$_{t+1}$[All cashflows from $t+1 \to$] ← Assuming alive at time $t+1$

$$\implies {}_tV = -P_t + e_t + (S_{t+1} + E_{t+1})\,A^{\,1}_{x+t:\overline{1}|} + A_{x+t:\overline{1}|}^{\,1}\,{}_{t+1}V$$

$$\implies ({}_tV + P_t - e_t)(1 + i_t) = (S_{t+1} + E_{t+1})\,q_{x+t} + p_{x+t}\,{}_{t+1}V$$

Interpretation: The cash at time t rolled up to time $t+1$ is sufficient to pay the **expected** amounts at time $t+1$, which are the death benefits (deaths) and policy values (survivors).

Policies with continuous cash flows – Thiele's Differential Equation:

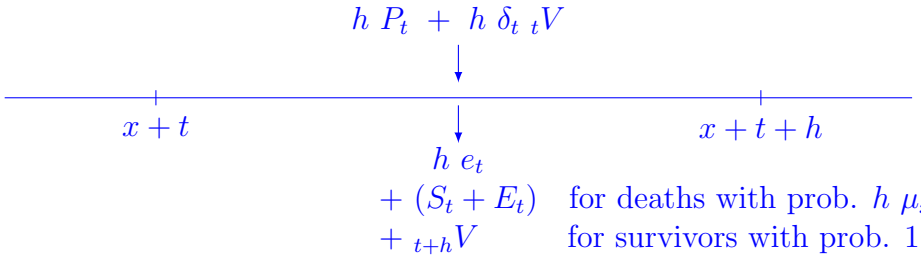
Consider a policy issued to a life (x) where cash flows occur continuously. Suppose this policy has been in force for t years, where t need not a non-negative integer. Let:

- P_t denote the rate per year of premium payable at time t,
- e_t denote the rate per year of premium-related expense payable at time t,
- S_t denote the sum insured payable immediately at time t if the policyholder dies,
- E_t denote the expense of paying the sum insured at time t,
- ${}_tV$ denote the gross premium policy value for a policy in force at time t

Let μ_{x+t} denote the force of mortality for the policy holder at age $x+t$ and let δ_t denote the force of interest assumed earned at time t. The quantities $e_t,\ E_t,\ \mu_{x+t}$ and δ_t are all as assumed in the policy value basis.

Let $0 \le t < t+h$, where h is small, and consider the EPV of future cash flows from $t \to t+h$ and from $t+h \to$:

$$ {}_tV \;=\; -\ {}_tV\ \delta_t\ h \;-\; (P_t - e_t)h \;+\; \mu_{x+t}\ h(S_t + E_t) \;+\; (1 - \mu_{x+t}\, h)\ {}_{t+h}V $$



$$ \Longrightarrow\quad {}_{t+h}V - {}_tV \;=\; h\left({}_tV\ \delta_t \;+\; P_t \;-\; e_t \;-\; \mu_{x+t}\ (S_t \;+\; E_t \;-\; {}_{t+h}V)\right) $$

$$ \Longrightarrow\quad \frac{d}{dt}\,{}_tV \;=\; {}_tV\ \delta_t \;+\; P_t \;-\; e_t \;-\; \mu_{x+t}\ (S_t \;+\; E_t \;-\; {}_tV) $$

Recursive calculation of policy values:

Example 6 *Consider a 20−year endowment insurance issued to a life aged 30. The sum insured, £100,000, is payable at the end of the year of death and level annual premiums are payable throughout the term. The basis for calculating premiums and policy values is:* } Annual cashflows

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$, *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075.$$

Interest: $\delta = 0.04$ *per year.* $\quad i = e^{0.04} - 1$

Expenses: Nil.

(a) Calculate the level annual premium, P.

(b) Explain why:

$$_{19}V = 100,000\, e^{-0.04} - P$$

(c) Calculate $_tV$ *for* $t = 19, 18, \ldots, 1, 0$.

Solution:

(a)

$$P\ \ddot{a}_{30:\overline{20}|} = 10^5\ A_{30:\overline{20}|}$$

Spreadsheet

$$\ddot{a}_{30:\overline{20}|} = \sum_{k=0}^{19} v^k\ {}_kp_{30} = 13.47634$$

$$A_{30:\overline{20}|} = 1 - d\ \ddot{a}_{30:\overline{20}|} = 0.47159$$

$$\Longrightarrow \quad P = £3,499.36$$

(b) The policy value for a policy in force just before time 20 $= 100,000$.

$$\Longrightarrow \quad (\,{}_{19}V + P)\ e^{0.04} = 10^5\ q_{49} + 10^5\ p_{49} = 10^5$$

(c)

$$_{19}V = 10^5\ e^{-0.04} - P = £92,579.58$$

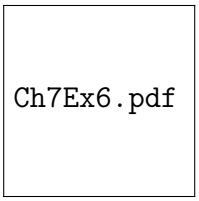
$$_{18}V = e^{=0.04}\left[10^5\ q_{48} + p_{48}\ {}_{19}V\right] - P = £85,533.63$$

$$\vdots \qquad \vdots$$

$$_0V = 0 \text{ since premium basis } = \text{ policy value basis}$$

and we calculated the premium using the equivalence principle.

$$\Rightarrow \text{ EPV[Premiums] } = \text{ EPV[Benefits] at time 0}$$


Ch7Ex6.pdf

Euler's method: Suppose that a function $u(t)$ is unknown. Suppose we know $u'(t) = \frac{du}{dt}$ for all t and we also know the initial condition $u(0)$, but we don't know how to integrate $u'(t)$. Then we estimate $u(t)$ by the following scheme:

$$\begin{aligned} u(0.1) &= u(0) + 0.1 \times u'(0) \\ u(0.2) &= u(0.1) + 0.1 \times u'(0.1) \\ &\vdots \end{aligned}$$

Example 7 (Euler's Method) *Consider a* $20-$*year endowment insurance issued to a life aged* 30. *The sum insured,* £100 000, *is payable* **immediately** *on death, or at the end of the term, and level premiums are payable* **continuously** *throughout the term. The basis for calculating premiums and policy values is:* } Continuous cashflows

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$, *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075.$$

Interest: $\delta = 0.04$ *per year.*

Expenses: Nil.

(a) *Calculate the annual amount of the premium,* P.

(b) *Explain why:* $\lim_{t \to 20^-} {}_tV = 100,000$.

(c) *Explain why:* *(for small* h*)*

$$\begin{aligned} {}_{t+h}V &\approx {}_tV + h({}_tV\,\delta + P - \mu_{30+t}(100,000 - {}_tV)) \qquad 0 \le t < t+h \le 20 \\ {}_{t-h}V &\approx {}_tV - h({}_tV\,\delta + P - \mu_{30+t}(100,000 - {}_tV)) \qquad 0 \le t-h < t \le 20 \end{aligned}$$

(d) *Calculate* ${}_{20}V$ *for* $t = 19.95,\ 19.9,\ 19.85,\ \cdots,\ 0.05,\ 0$.

Solution: $\int_0^{20} e^{-\delta t}\ {}_tp_{30}\ dt = 13.16986$

(a) $P\ \bar{a}_{30:\overline{20}|} = 10^5\ \bar{A}_{30:\overline{20}|} \to P = £3,593.10 > 3,499.36$ ($1 - \delta\ \bar{a}_{30:\overline{20}|}$)

(b) As $t \to 20$ the EPV of future $\begin{cases} \text{benefits} & \to 10^5 \\ \text{premiums} & \to 0 \end{cases}$.

(c) Thiele's Differential Equation

$$\frac{d}{dt}\,{}_tV = {}_tV\,\delta + P - \mu_{30+t}\left(10^5 - {}_tV\right)$$

$$\text{Small } h: \quad \frac{d}{dt}\,{}_tV \cong \frac{{}_{t+h}V - {}_tV}{h} \cong {}_tV\,\delta + P - \mu_{30+t}\left(10^5 - {}_tV\right)$$

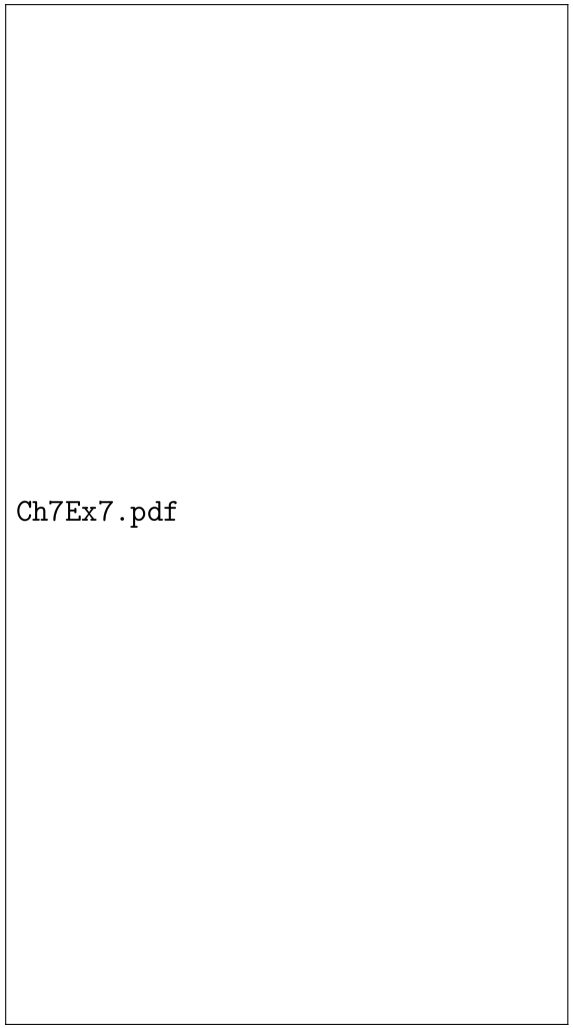
$$\cong \frac{{}_{t-h}V - {}_tV}{-h} \cong {}_tV\,\delta + P - \mu_{30+t}\left(10^5 - {}_tV\right)$$

(d) See spreadsheet

Results: $\left.\begin{aligned} h = 0.05 &\to {}_0V = -64.72 \\ h = 0.025 &\to {}_0V = -32.34 \end{aligned}\right\} \to$ Should be 0

$${}_{t-h}V \cong {}_tV - h\,[\,{}_tV\delta + P - \mu_{30+t}\,(10^5 - {}_tV)]$$

Step size
$= 0.1$



20 100 000 100 000

Note: Since policy value basis = premium basis **and** the premiums are calculated using the equivalence principle $\Longrightarrow$ we know that ${}_0V = 0$. We could have calculated

$$ {}_0V \to {}_1V \to {}_2V \to \cdots \to {}_{19}V \to {}_{20}V \qquad \text{(Ex. 6)} $$
$$ {}_0V \to {}_hV \to {}_{2h}V \to \cdots \to {}_{20-h}V \to {}_{20}V \qquad \text{(Ex. 7)} $$

But this is not possible in general.

Comments on recursive calculations and Euler's Method:

- The recursive calculation of policy values is very flexible: we can incorporate variable benefits, expenses, premiums, interest rates.

  GOOD :)

- Euler's method is very crude.

  NOT GOOD :(

  Using $h = 0.05$, ${}_0V = -64.72$, should be 0.

- Example 7: smaller values of h are likely to give better approximations.

  $h = 0.025 \quad \Longrightarrow \quad {}_0V = -32.34$

- There are better methods than Euler's to obtain numerical solutions to differential equations, *e.g.* a Runge–Kutta Method.

  See any textbook on numerical analysis.

Asset shares:

- The **policy value** at duration t represents the cash the insurance company **needs** at time t in respect of a policy still in force.

  ↑ To meet future liabilities.

- The **asset share** at duration t represents the cash the insurance company **actually has** at time t in respect of a policy still in force.

- The asset share is the accumulation to time t of past premiums received minus past claims and past expenses paid.

- The asset share is calculated assuming the policy being considered is one of a large number, N, of identical policies issued simultaneously:

  $$\text{Asset share} \quad = \quad \text{total accumulated funds/number of survivors}$$

  (which does not depend on N).

Note the three IFs

- **If** the premium is calculated using the equivalence principle, **if** the premium basis $\equiv$ the policy value basis and **if** the past experience of the group of policies (survival, interest, expenses, bonus) is as assumed in this common basis, then:

  $$\textbf{Asset share} \quad = \quad \textbf{Policy value}$$

Example 8

A life office sells a 10 *year endowment policy to a life aged exactly* 60.

The sum insured of £100,000 *is payable at the end of the year of death or on survival to age* 70.

Level premiums are payable annually in advance throughout the term of the policy.

Premium and policy value basis:

Survival model: The force of mortality is given by $\mu_x = A + Bc^x$ *where:*

$$A = 0.0001; \quad B = 0.00035; \quad c = 1.075.$$

Interest: 6% *per year.*

Expenses: 10% *of each premium.*

(a) Calculate the annual premium. ← Equivalence principle

$$0.9\, P\, \ddot{a}_{60:\overline{10}|} = 10^5\, A_{60:\overline{10}|} = 10^5\,(1 - d\, \ddot{a}_{60:\overline{10}|})$$

$\ddot{a}_{60:\overline{10}|} \uparrow$ 6.84512 ← See spreadsheet for Ch.7

$$\Longrightarrow \quad P = £9,942.87$$

(b) Calculate the gross premium policy value at the start of the 4*-th year.* (Duration 3 years)

$$_3V = 10^5\, A_{63:\overline{7}|} - 0.9\, P\, \ddot{a}_{63:\overline{7}|}$$

$\ddot{a}_{63:\overline{7}|} \uparrow$ 5.3281

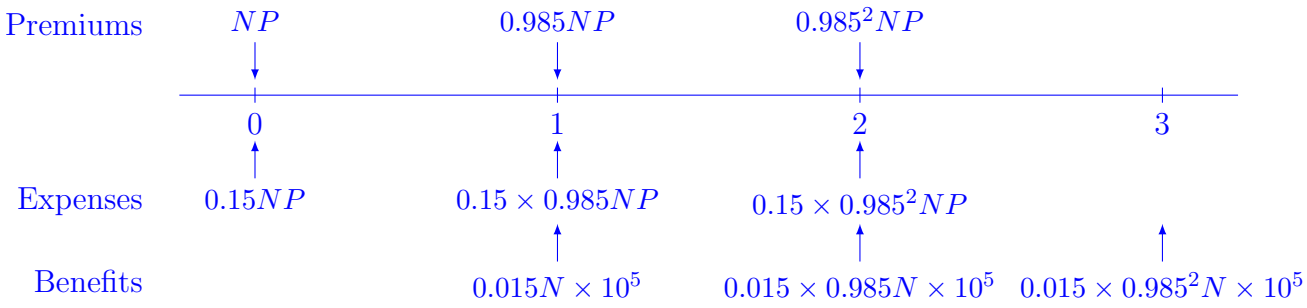
$$\Longrightarrow \quad _3V = £22,162.30$$

(c) *Calculate the asset share at the start of the 4-th year assuming that in the first 3 years:*

Actual experience or experience basis

- *Interest has been 7.5% per year*
- *The rate of mortality (for similar policies) has been* $q_x = 0.015$
- *Expenses were 15% of the premiums in the first three years.*

Consider N identical policies issued simultaneously.



Accumulation to $t = 3$

$$NP(1 - 0.15)\left[1.075^3 + 0.985(1.075)^2 + 0.985^2(1.075)\right] - 0.015 \times 10^5 N\left[1.075^2 + 0.985(1.075) + 0.985^2\right]$$

$$= £24,157 \times N$$

No. of survivors at $t = 3 = 0.985^3 N$

$$\Longrightarrow \text{Asset share} = \frac{24\,157N}{0.985^3 N} = £25,278$$

Note: The value of N does not matter.

$$\text{Asset share} = 25,278 > 22,162 = {}_3V$$

$$\text{Actual interest} = 7.5\% > 6\% \quad \text{expected} \quad \text{GOOD :)}$$

$$\text{Actual expenses} = 15\%P > 10\%P \text{ expected} \quad \text{NOT GOOD :(}$$

$$\text{Actual mortality} = 0.015 < q_{60} \quad \text{expected} \quad \text{GOOD :)}$$

↑

i.e. assumed in premium basis

We have assumed for convenience that interest and expenses are constant each year

(d) *Show that the asset share at the start of the 4-th year can be written:*

$$\left((P-E)\ddot{a}^*_{60:\overline{3}|} \;-\; 10^5 A^{*1}_{60:\overline{3}|}\right) \;/\; A^{*\;\frac{1}{}}_{60:\overline{3}|}$$

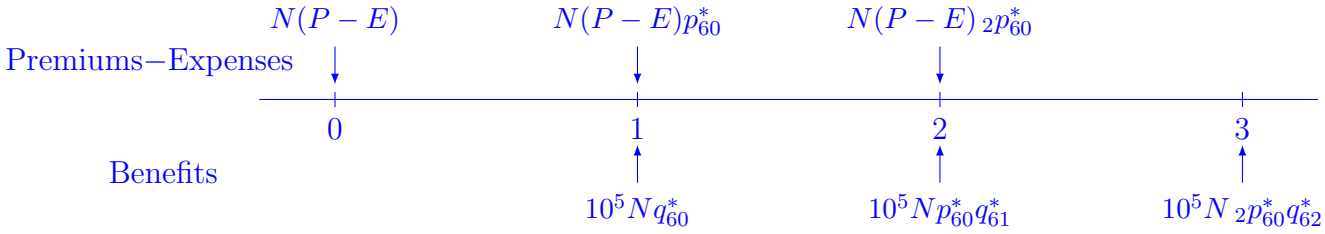
assuming that in the first 3 years:

Interest has been i^* per year

The rate of mortality (for similar policies) has been according to some given survival model

Expenses per policy have been E in the first 3 years.

where * denotes calculation at rate i^* using the survival model actually experienced.



Asset Share:

$$\frac{\not{N}\left\{(P-E)\left[1+v^*\,p^*_{60}+(v^*)^2\;{}_2p^*_{60}\right]-10^5\left[v^*\,q^*_{60}+(v^*)^2\;p^*_{60}q^*_{61}+(v^*)^3\;{}_2p^*_{60}q^*_{62}\right]\right\}}{\not{N}\;{}_3p^*_{60}\;(v^*)^3}$$

$$= \frac{(P-E)\;\ddot{a}^*_{60:\overline{3}|} \;-\; 10^5\;A^*_{60:\overline{3}|}}{A^{*\;\frac{1}{}}_{60:\overline{3}|}}$$

(e) *Show that*
if *the experience has been as in the premium basis ($E = 0.1P$; $i^* = 0.06$ and the experienced survival model $\equiv$ the model in the premium basis) and*
if *the policy value basis $\equiv$ the premium basis,*
then the **asset share = policy value.**

Assumption: $E = 0.1P$, $i^* = 0.06$, ${}_tp_{60}$ as in premium & policy value basis.

$$\Longrightarrow \text{ Asset share } = \frac{0.9\;P\;\ddot{a}_{60:\overline{3}|} \;-\; 10^5\;A^1_{60:\overline{3}|}}{A_{60:\overline{3}|}^{\;\frac{1}{}}}$$

Equation of value: $0.9\;P\;\ddot{a}_{60:\overline{10}|} \;=\; 10^5\;A_{60:\overline{10}|}$

$$\Longrightarrow \quad 0.9\;P\left[\ddot{a}_{60:\overline{3}|} \;+\; A_{60:\overline{3}|}^{\;\frac{1}{}}\;\ddot{a}_{63:\overline{7}|}\right] \;=\; 10^5\left[A^1_{60:\overline{3}|} \;+\; A_{60:\overline{3}|}^{\;\frac{1}{}}\;A_{63:\overline{7}|}\right]$$

$$\Longrightarrow \quad \frac{0.9\;P\;\ddot{a}_{60:\overline{3}|} \;-\; 10^5\;A^1_{60:\overline{3}|}}{A_{60:\overline{3}|}^{\;\frac{1}{}}} \;=\; 10^5\;A_{63:\overline{7}|} \;-\; 0.9\;P\;\ddot{a}_{63:\overline{7}|}$$

↑ Asset share ↑ Policy value

Annual Profit:

Policy duration ↓

Consider a large number N of identical life policies in year $t \to t+1$, where t is an integer.

Sum insured S payable at the end of the year of death.

Annual premiums P.

Policy value basis: interest at rate i per year; expenses e at t; survival probability p_{x+t}. Then:

↙ Recursive formula with annual cashflows and policy value basis

$$N({}_tV + P - e)(1 + i) = N(S\, q_{x+t} + p_{x+t}\, {}_{t+1}V)$$

Interpretation: **IF** the assumptions (interest, expenses, survival model) for this year turn out to be correct, then:
policy value + premiums − expenses at t will be exactly enough to pay benefits at $t+1$ and set up the policy values required at $t+1$ for the next year. i.e. no profit or loss during the year.

IF the assumptions turn out to be wrong (almost always):
the insurance company may have too much (surplus/profit)
or too little (deficit/loss)
to set up the required policy values after paying claims for the year.

Profit Sources:

Annual profit > 0 if

"expected" means according to the policy value basis ↙

- Interest higher than expected
- Expenses lower than expected
- Fewer policyholders die than expected (insurances)
- More policyholders die than expected (annuities)

Annual Profit from Mortality, Expenses and Interest:

Consider N insurance policies (as above)

Then:

$$N({}_tV + P - e)(1 + i) = N(S\, q_{x+t} + p_{x+t}\, {}_{t+1}V) \quad \text{(as above)}$$

Suppose now:

The actual expenses are e^* per policy at the start of the year.
Interest is earned at rate i^* per year in the year.
n_d policyholders die in the year. n_d : integer ≥ 0

Then the cash available at the end of the year is:

$$N({}_tV + P - e^*)(1 + i^*)$$

and the cash required at the end of the year is:

$$n_d\, S + (N - n_d)\, {}_{t+1}V$$

Hence the profit in the year from mortality, expenses and interest is:

$$N({}_tV + P - e^*)(1 + i^*) - n_d\, S - (N - n_d)\, {}_{t+1}V$$

Do not remember this formula! Remember the method.

Special case – Mortality profit:

Suppose $i^* = i$ and $e^* = e$ (interest and expenses as expected $\rightarrow$ the only source of profit/loss is mortality). Then, the (mortality) profit in the year is:

$$N({}_tV + P - e^*)(1 + i^*) - n_d\, S - (N - n_d)\, {}_{t+1}V$$

Recursive equation

LHS $= \underbrace{N({}_tV + P - e)(1 + i)}$ $- n_d\, S - (N - n_d)\, {}_{t+1}V$

RHS $= \underbrace{N\, p_{x+t}\, {}_{t+1}V + N\, q_{x+t}\, S}$ $- n_d\, S - (N - n_d)\, {}_{t+1}V$

$$= (N\, q_{x+t} - n_d)(S - {}_{t+1}V)$$

compare with Thiele's equation

- $S - {}_{t+1}V$ is the **death strain at risk** for the policy in the year (= the extra cost of providing the sum insured compared to the policy value at the end of the year).
- $Nq_{x+t}(S - {}_{t+1}V)$ is the **expected death strain at risk**
- $n_d(S - {}_{t+1}V)$ is the **actual death strain at risk**
- $Nq_{x+t} - n_d$ is the expected number of deaths minus the actual number.
- For an annuity policy, S will be zero and the death strain at risk is negative.

$\Longrightarrow$ Profit: Expected death strain at risk $-$ Actual death strain at risk.

Example 9 *On 1 January 1994 a life office issued a number of identical whole life insurances to lives then aged* 55. *Each policy had a sum insured of* £50,000, *payable at the end of the year of death, and annual premiums payable for at most* 10 *years.*

The premium and gross premium policy value basis is as follows:

Tables! { *Mortality: AM92 Select.* ← *at age* 55
Interest: 4% *per year.*
Expenses: 10% *of each gross premium.*

(a) *Calculate the gross annual premium,* P.

(b) *On 1 January 2002 there were* 200 *of these policies still in force. In 2002:* 5 *policyholders died.*
The actual expenses on 1 January 2002 were 12% *of each gross premium.*
The actual rate of interest earned in 2002 was $6\frac{1}{4}\%$.
Calculate the office's profit in 2002.

(c) *On 1 January 2008 there were* 180 *of these policies still in force. In 2008:* 4 *policyholders died.*
The actual rate of interest earned in 2008 was $5\frac{1}{2}\%$.
Calculate the office's profit in 2008.

Solution:

(a) $0.9\ P\ \ddot{a}_{[55]:\overline{10}|} = 50,000\ A_{[55]} \rightarrow P = £2,625.11$

(b)

$$ {}_8V = 50,000\ A_{63} - 0.9 \times 2,625.11 \times \ddot{a}_{63:\overline{2}|} = £20,335.57 $$

$$ {}_9V = 50,000\ A_{64} - 0.9 \times 2,625.11 \times \ddot{a}_{64:\overline{1}|} = £23,303.90 $$

$$ \text{Profit in 2000} : 200\ [20,335.57 + 2,625.11 - 0.12 \times 2,625.11]\,(1.0625) - 5 \times 50,000 - 195 \times 23,303.90 $$

$$ = £17,944 \qquad \text{Note } 200\ q_{63} = 2.27 \begin{cases} \text{Mortality loss} \\ \text{Expense loss} \\ \text{Interest profit} \end{cases} $$

(c)

$$ {}_{14}V = 50,000\ A_{69} = £29,319.00 $$

$$ {}_{15}V = 50,000\ A_{70} = £30,048.50 $$

$$ \text{Profit in 2008} : 180 \times 29,319.00 \times 1.055 - 4 \times 50,000 - 176 \times 30,048.50 $$

$$ = £79,142 \qquad \text{Note } 180\ q_{69} = 4.001 \begin{cases} \text{Mortality very small profit} \\ \text{Expense no contribution} \\ \text{Interest profit} \end{cases} $$

Policy Alterations :

- Reducing the premium payable
- No more premiums payable → **paid-up policy (PUP)** with a **paid-up sum insured**
- Changing the policy term
- Changing a policy from a whole life insurance to an endowment insurance
- Changing a policy from with profit to without profit
- Cancelling (= lapsing or surrendering) a policy and accepting a reduced sum insured now (= surrender value)
- .....

Note: Alterations are **always** at the request of the policyholder.

Basic method for calculating altered premiums/benefits :

$$ {}_tV \;=\; {}_tV^{*} \;+\; C $$

where:

(a) ${}_tV$ is the policy value **before** making the alteration.

(b) ${}_tV^{*}$ is the policy value **after** making the alteration.

(c) C is the cost to the insurer of making the alteration.

(d) The **basis** for these calculations needs to be specified. Need not be the same as any other basis.

(e) If the basis is realistic, this method implies the insurer expects to make neither a profit nor a loss from the alteration.

(f) This is the method to be used unless told otherwise.

Example 10 *A* 20*-year annual premium endowment insurance is issued to a life aged* 40*. Annual premiums of* £500 *are payable throughout the term and the sum insured of* £15,000 *is payable at the end of the year of death or at maturity.*
Immediately before the 10*-th premium is due the policyholder requests that no more premiums are paid.*

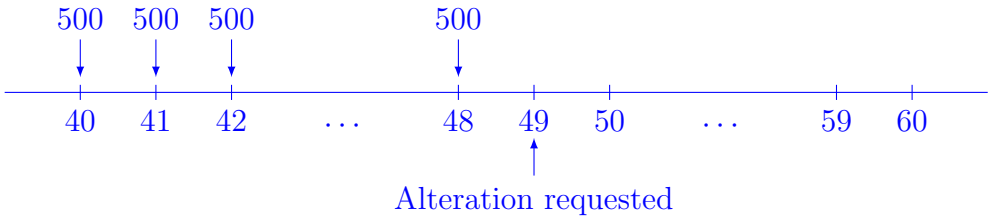
(a) Calculate the surrender value payable immediately.

(b) Calculate the paid–up sum insured.

The basis for calculating revised benefits is:
Tables! { *Survival model: AM92 ultimate*
Interest: 6% *per year*
Expenses: 10% *of each premium payable and* £100 *for making the alteration.*

Solution:

Premiums



Policy value before the change:

$$15,000\ A_{49:\overline{11}|} \ - \ 0.9 \ \times \ 500\ \ddot{a}_{49:\overline{11}|} \ = \ £4,293$$

(a) Surrender value (SV) is paid now (= at age 49)

$$4,293 \ = \ \text{SV} \ + \ 100 \quad \Longrightarrow \quad \text{SV} \ = \ £4,193$$

(b) The policy value after the change is $S \cdot A_{49:\overline{11}|}$ where

S = the paid-up sum insured

$$\Longrightarrow \quad 4,293 \ = \ S \cdot A_{49:\overline{11}|} \ + \ 100$$

$$\Longrightarrow \quad S \ = \ £7,860$$